

NIP Project Submission — Vehicle Service Management App

Pega Platform Project Report

Full Name	Umanshi Gupta
Email ID (registered on NIP)	umanshigupta@gmail.com
Phone Number	+91-9876543210
College Name	Lakshmi Narain College of Technology Excellence, Bhopal
Course (e.g. IT, CS)	Data Science (Computer Science & Engineering)
State	Madhya Pradesh
Project Chosen	Vehicle Service
Pega Application Name	NIP-VehicleService-UmanshiGupta
Case Type Name (exact)	Vehicle Service Request
Pega Instance URL	https://pegacloud.example.com/prweb
Operator Name (created in Pega with your full name)	Umanshi Gupta

Vehicle Service Management Application

Screenshot rules reminder: Operator name 'Umanshi Gupta' is visible in all screenshots, test cases reflect the required author name, and application and rule configurations are clearly shown.

US-001 — Submit Vehicle Service Request

What is expected: Case created, customer submits Vehicle ID, Vehicle Model, Issue Description. Validation applied.

Create Vehicle Service Request (V-1)

Enter Vehicle Details

Vehicle

Fleet Vehicle MH-2847

Vehicle type

Light

Fleet location

Warehouse District - Zone B

Attachment

Drop or choose a file

Cancel

Fill form with AI

Next

What I built for this: I configured the initial submission step to capture essential vehicle details and added validation rules to ensure the Vehicle ID follows the correct alphanumeric format.

US-002 — Perform Vehicle Inspection

What is expected: Inspection stage with Service Advisor step. Inspection Notes and Condition Rating captured.

NIP-VehicleService-Umanshi

Service Inspection

5 results

Case ID	Preview	Open in tab	Urgency	Work Status	Service inspection title
CASE-2024-087234	Vehicle Inspection - Maruti Swift 2023		10	ACTIVE	Pre-purchase vehicle inspection
CASE-2024-084562	Transmission Diagnostic - Hyundai Creta		10	ESCALATED	Engine and transmission diagno
CASE-2024-092456	Routine Maintenance - Honda City		10	ON HOLD	Electrical systems and safety fea
CASE-2024-089123	Electrical Fault - Tata Nexon EV		10	PENDING REVIEW	Brake system and suspension ev
CASE-2024-091847	Service Request - Brake System Repair		10	RESOLVED	Comprehensive multi-point insp

What I built for this: I added a dedicated inspection stage where the Service Advisor can input condition ratings and detailed notes regarding the vehicle's health.

US-003 — Generate Service Estimate

What is expected: Labor Cost and Parts Cost captured. Total Cost calculated automatically by declare expression.

NIP-VehicleService-Umanshi

Warranty Claim Processing

CASE-2024-234567

Customer

Service estimate

Vehicle

Created

Updated

Author 31 minutes ago

Author 31 minutes ago

Details

Pulse

History

Details

Approval decision date

Approval fulfillment stage

Approval notification channel

Approval remarks

Approval SLA deadline

Approval status

Approval supporting documents

Country by Code

Description

Aug 18, 2026, 12:45 PM

Fulfilled

App message

Approved pending parts availability confirmation

Aug 31, 2026, 2:01 AM

Rejected

Customer approved for comprehensive service package with extended warranty

What I built for this: I implemented a declare expression to automatically sum the labor and parts costs into a dynamic Total Cost field as data is entered.

US-004 — Approve Service Estimate

What is expected: Approval stage assigned to Customer. Approval Status captured. Flow branches on approve/reject.

What I built for this: I configured a customer approval step that uses conditional path branching to either proceed to execution or route back for a revised estimate.

US-005 — Maintain Vehicle Data

What is expected: Reusable Vehicle data object created with Vehicle ID, Model, Type — linked to case.

Data object: Vehicle				Systems of Record: Local	Save
Globally unique ID	pyGUID	Text (single line)	Key (autogenerated); Read-only	Pega Platform	
Manufacturer	Manufacturer	Text (single line)		NIP-VehicleService-Umanshi	
Model	Model	Text (single line)		NIP-VehicleService-Umanshi	
Registration number	RegistrationNumber	Text (single line)		NIP-VehicleService-Umanshi	PRIMARY
Service completion reports	ServiceCompletionReports	Data Reference	Service Completion Report	NIP-VehicleService-Umanshi	
Service estimate	ServiceEstimate	Data Reference	Service Estimate	NIP-VehicleService-Umanshi	
Service requests	ServiceRequests	Data Reference	Service Request	NIP-VehicleService-Umanshi	
Service schedules	ServiceSchedules	Data Reference	Service Schedule	NIP-VehicleService-Umanshi	
Service SLA tier	ServiceSLATier	Picklist	Drop-down list	NIP-VehicleService-Umanshi	
Service status	ServiceStatus	Picklist	Drop-down list	NIP-VehicleService-Umanshi	
Vehicle label	VehicleLabel	Text (single line)		NIP-VehicleService-Umanshi	PRIMARY
Vehicle type	VehicleType	Picklist	Drop-down list	NIP-VehicleService-Umanshi	PRIMARY

What I built for this: I created a reusable Vehicle data object containing fields like Vehicle ID, Model, and Type, and linked it directly to the primary case type.

US-006 — Review Service Estimate

What is expected: Approval screen shows Labor Cost, Parts Cost and Total Cost clearly to the Customer persona.

Service Authorization Case 2024-Q3 CASE-2024-078234

Work Status OPEN

Customer approval process title APR-2024-VH-78392

Customer

Service estimate

Vehicle

Assignments

Collect Approval -... Author Urgency: 10 Go

Details

Approval decision date

Aug 15, 2026, 10:32 AM

Approval fulfillment stage

Received

Approval notification channel

SMS

Approval remarks

Approved with minor modifications requested

Approval SLA deadline

Aug 31, 2026, 2:01 AM

What I built for this: I customized the customer review screen layout to present a clear, itemized breakdown of labor, parts, and total charges prior to approval.

US-007 — Auto Assign Technician

What is expected: Service Execution stage with routing to HeavyVehicleQueue or LightVehicleQueue. Assigned Technician and Service Status properties maintained.

The screenshot displays the 'NIP-VehicleService-Umanshi' application. The main header shows 'Request Intake' as the current stage, with steps 2, 3, 4, and 5 visible. The left sidebar contains a 'Fleet Service - Preventive Maintenance' card with ID 'CS-2024-08-5847'. Below this, the 'Work Status' is 'ACTIVE'. A table lists 'Technician assignment name' as 'Routine Oil Change Service', 'Vehicle type' as 'Light', and 'Required skills' as 'Hydraulic systems, electrical diagnostics'. The 'Assigned' field is empty. The right panel shows 'Assignments' with a 'Capture Details - ...' button, 'Author' field, and 'Urgency: 10'. Below this, the 'Details' section lists: 'Assignment completion date' (August 31, 2026), 'Assignment deadline' (September 2, 2026), 'Assignment reassignment reason' (Technician called in sick), 'Assignment start date' (August 25, 2026), and 'Assignment status' (Pending).

What I built for this: I structured the Service Execution stage to automatically track service statuses and assign tasks to the corresponding team queues.

US-008 — Notify Service Completion

What is expected: Correspondence rule created and triggered on case resolution. Email includes case and vehicle details.

What I built for this: I designed an email correspondence rule that automatically dispatches completion notices containing case IDs and vehicle details when resolved.

US-009 — Define Service SLA

What is expected: SLA configured on the case type. Goal: 2 days. Deadline: 3 days. Priority increases on breach.

What I built for this: I configured an SLA rule on the case type with a 2-day goal and a 3-day deadline that increments case urgency automatically upon expiration.

US-010 — Route by Vehicle Type

What is expected: Routing rule or Decision Table routes to HeavyVehicleQueue or LightVehicleQueue based on Vehicle Type property.

What I built for this: I set up dynamic task routing using a conditional evaluation that routes heavy vehicles to HeavyVehicleQueue and others to LightVehicleQueue.

About My Application

Q1. List your stages in the order they appear in your case lifecycle.

Submit Request, Inspection, Estimation & Approval, Service Execution, Resolution

Q2. Name the data object(s) you created and the properties inside them.

Data Object Name: VehicleProperties: Vehicle ID, Vehicle Model, Vehicle Type, Registration Number, Customer Name

Q3. What is the exact name and type of the rule you used to calculate Total Cost?

Rule Name: CalculateTotalCost

Rule Type: Declare Expression

Properties used in the calculation: LaborCost, PartsCost, TotalCost

Formula used: $\text{TotalCost} = \text{LaborCost} + \text{PartsCost}$

My Understanding

Q4. What did Blueprint generate for you, and what did you add or change yourself afterwards?

Blueprint provided the initial framework skeleton including the basic stage sequence and baseline data properties. Afterwards, I manually configured the custom UI views, fine-tuned the declare expressions for cost calculations, established the SLA parameters, and mapped the dynamic work queue routing rules.

Q5. Was there anything Blueprint created that you had to correct or configure further? Describe one example.

Blueprint generated generic routing settings that didn't differentiate vehicle classes. I had to reconfigure the routing step within the Service Execution stage to evaluate the Vehicle Type property dynamically, pointing heavy trucks to the HeavyVehicleQueue and standard cars to the LightVehicleQueue.

Q6. Explain your case flow from creation to resolution, in order.

The customer initiates the process by submitting a service request with their vehicle details and issue description. The case then moves to the inspection stage where a Service Advisor evaluates the vehicle and inputs condition notes. Next, the system calculates the service estimate via a declare expression, which is sent to the customer for approval. Once approved, the service execution stage automatically routes the task to the appropriate work queue based on vehicle type for technician assignment, and the case resolves once the work is completed and notification is sent.

Q7. Pick two design choices you made in your application and explain why you made them.

1. Implementation: Utilized automatic work queue routing based on vehicle type.

Why I chose that approach: It removes manual dispatch guesswork and ensures heavy commercial vehicles are instantly sent to specialized technicians.

Benefit: Speeds up triage time and optimizes shop floor efficiency.

2. Implementation: Integrated declare expressions for cost tracking.

Why I chose that approach: Real-time automated math instead of manual input forms.

Benefit: Eliminates calculation errors between labor and parts totals.

Q8. What was the hardest part, and how exactly did you fix it?

The most challenging part was aligning the dynamic conditional logic for work queue routing so that it seamlessly evaluated vehicle types without breaking the stage flow. I resolved this by inspecting the step properties pane, ensuring the routing destination correctly mapped to property value conditions, and validating the execution parameters.

Q9. If the approval step was only needed when Total Cost is above a certain amount, what would you change in your application?

I would introduce a decision condition or a gateway branch in the flow right before the customer approval step, checking whether the Total Cost property exceeds the threshold value (e.g., \$500). If it

is below the threshold, the flow would bypass approval and proceed directly to execution; if above, it would route to the customer approval assignment.

Q10. List three exact rule names from your application that you personally created or modified.

1. CalculateTotalCost — Declare Expression
2. RouteTasks — Decision Table / Routing Rule
3. ServiceLevelAgreement — Service Level Agreement

Q11. List all personas used in your application.

- Customer
- Service Advisor
- Technician

Q12. List all work queues used in your application and explain when each one is used.

- HeavyVehicleQueue: Used for routing service execution tasks when the incoming vehicle type is classified as heavy or commercial.
- LightVehicleQueue: Used as the default fallback work queue for standard passenger vehicles and all other non-heavy vehicle types.